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SUPERSEDING
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FEDERAL SPECIFICATION

DETERGENT, GENERAL PURPOSE, SOLVENT TYPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers solvent-type, liquid detergent for heavy duty cleaning applications where grease, oil, and carbonaceous deposits are a problem (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

O-S-642	-	Sodium Phosphate, Tribasic, Anhydrous; Dodecahydrate; and Monohydrate; Technical.
QQ-S-766	-	Steel, Plates, Sheets and Strip, Corrosion-Resisting.
SS-T-312	-	Tile, Floor: Asphalt, Rubber, Vinyl, Vinyl-Asbestos.
TT-E-508	-	Enamel: Interior, Semi-Gloss, Tints and White.
TT-E-543	-	Enamel, Undercoat, Interior, Tints and White.
LLL-F-1238	-	Floor Covering, Linoleum.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter.
PPP-F-320	-	Fiberboard, Corrugated and Solid, Sheet Rock (Container Grade), and Cut Shapes.
PPP-P-704	-	Pails, Metal (Shipping, Steel, 1 Through 12 Gallon).
PPP-P-1655	-	Pail, Plastic, Shipping and Storage.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 536/GEN - Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing.
Fed. Test Method 536/6701-T - Cleaning Efficiency Test.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

16 CFR 1500 - Federal Hazardous Substances Act.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials(ASTM) Standards:

D 95 - Water in Petroleum Products and Bituminous Materials by Distillation.
D 322 - Dilution of Gasoline - Engine Crankcase Oils.
D 523 - Specular Gloss.
D 820 - Chemical Analysis of Soaps Containing Synthetic Detergents.
D 1310 - Flash Point of Liquids by Tag Open-Cup Apparatus.

(Application for copies, should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Soap and Detergent Association:

Test Procedure and Standards - ABS and LAS Biodegradability; Scientific and Technical Report No. 3.

(Application for copies should be addressed to the Soap and Detergent Association, 475 Park Avenue South, New York, NY 10016.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The product shall be a uniform liquid containing surfactants, suitable solvents, and builders to provide a satisfactory cleaner for use in heavy duty general maintenance work. The product shall be safe for use by personnel without gloves or other protective equipment, shall strip heavy soil rapidly and keep the soil suspended in the cleaning solution to such a degree that it is readily removed by mop, squeegee or vacuum pick-up without leaving any appreciable deposit on the surface. The product shall be clear and free of sediment at time of delivery.

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3.2 Odor. The product shall have no objectionable odor, nor shall it develop an objectionable odor in water solutions, in use, or when stored in the original container.

3.3 Stability in storage. The product shall be stable and not lose its original effectiveness or otherwise deteriorate when stored in a closed shipping container for a period of not less than one year after receipt. The product shall show no tendency to separate while in storage under extreme ranges of temperature (-2° to 50°). Unless otherwise specified, the contractor shall furnish a certified laboratory report stating compliance with this requirement.

3.4 pH value. The pH value of a 1.0 percent by volume solution of the product with distilled water shall be not less than 9.0 or more than 10.5, when tested as specified in 4.4.1.

3.5 Solubility in hard water. The product shall form a stable uniform emulsion in synthetic hard water (300 ppm hardness), when tested as specified in 4.4.2.

3.6 Rinsing properties. The product shall be free-rinsing, when tested as specified in 4.4.3.

3.7 Cleaning efficiency. A solution of one-half percent by volume (one-half volume of product to 99.5 volumes of hard water) shall exhibit a cleaning efficiency of not less than 80 percent, when tested as specified in 4.4.4.

3.8 Composition. The product shall conform to the following requirements, when tested as specified in 4.4.5.

	Percent
Water	45 (max.)
Total solids	11 (min.)
Synthetic organic detergent	4 (min.)
Volatile organic solvents	36 (min.)

3.9 Deleterious action on painted surfaces. A 0.2 percent by volume solution of the product shall not cause greater than one-half of the loss of 60 degree specular gloss of painted surfaces caused by a 0.2 percent solution of trisodium phosphate, when tested as specified in 4.4.6.

3.10 Deleterious action on linoleum, rubber, and vinyl-asbestos tile. A 2.0 percent by volume solution of the product shall not cause any softening, swelling, cracking or whitening or bleaching of the test specimens, when tested as specified in 4.4.7.

3.11 Biodegradability. The synthetic organic detergent used in the manufacture of the product shall be 90 percent (minimum) biodegradable, when tested as specified in 4.4.8.

3.12 Flash point. The flash point of the product shall not be less than 90°C (194°F), when tested as specified in 4.4.9.

3.13 Labeling. The product shall be labeled in accordance with the Federal Hazardous Substances Act. The supplier's recommended uses, directions for use, and dilutions for use shall be clearly printed on the label. (For cleaning floors and general maintenance cleaning operations, the recommended dilution shall approximate 2 ounces per gallon; for washing walls, 1 ounce per gallon; for removal of oil, grease, and heavily encrusted soil, 5 ounces per gallon.) The following statement shall be included: DO NOT USE ON ASPHALT TILE.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein. For purposes of sampling, an inspection lot for acceptance inspection and tests shall consist of all material of the same type submitted for inspection and delivery at one time.

4.3 Inspection.

4.3.1 Inspection of material and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.3.2 Inspection and test of the end item.

4.3.2.1 Examination of the end item. The end item shall be examined in accordance with the classification of defects, inspection levels and acceptable quality levels (AQL's) set forth below. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled primary containers of product of the same type for examinations under 4.3.2.1.1 and 4.3.2.1.2, and in units of shipping containers for examination under 4.3.2.1.4.

4.3.2.1.1 Examination of the end item for defects in appearance of product. The sample unit for this examination shall be one open filled primary container. The inspection level shall be S-2 with an AQL of 2.5 percent defective.

Examine	Defect
Appearance of product	Presence of dirt or foreign matter. Not homogeneous. Form not as specified. Objectionable odor.

4.3.2.1.2 Examination of the end item container for defects in net contents. The sample unit for this examination shall be one filled container. The average net contents shall be not less than the specified or indicated quantity. The inspection level shall be S-2 with an AQL of 2.5 percent defective.

4.3.2.1.3 Testing of the end item. The end item shall be tested for the characteristics listed in table I. Sixteen ounces of material shall be drawn from each of five shipping containers and composited. The test sample shall consist of a 32-ounce composite. Unless otherwise specified, duplicate determinations shall be performed on the composite. The lot shall be unacceptable if the composite sample fails to meet any requirements referenced in table I.

TABLE I. Instructions for testing

Characteristic	Specification reference		Number of determinations per unit	Results reported as	
	Requirement	Test Method	Composite 1/	Pass or fail 2/	Numerically to nearest
pH value	3.4	4.4.1	Average of 2	--	0.1 pH
Solubility in hard water	3.5	4.4.2	2	X	--
Rinsing properties	3.6	4.4.3	2	X	--
Cleaning efficiency	3.7	4.4.4	Average of 3	--	0.1 percent
Composition	3.8	4.4.5	Average of 2	--	0.1 percent
Deleterious action on painted surfaces	3.9	4.4.6	2	X	0.1 percent decrease in gloss
Deleterious action on linoleum, rubber, and vinyl-asbestos tile	3.10	4.4.7	2	X	--
Biodegradability 3/	3.11	4.4.8	Average of 2	--	0.1 percent
Flash Point	3.12	4.4.9	Average of 3	--	1°C

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1/ Test report shall include all values on which results are based.
2/ If failure is indicated, report either description of failure or numerical point of failure, as applicable.
3/ The testing shall be performed and actual test data (field or laboratory) furnished on the first lot only. For each subsequent lot, a certificate shall be furnished indicating compliance with this requirement. If at any time the synthetic detergent is changed, the certificate of compliance shall be accompanied with test data as for the first lot.

4.3.2.1.4 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table II. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

TABLE II. Examination of preparation for delivery

Examine	Defect
Contents	Not as specified.
Containers	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers.
	Cushioning inadequate, improper or missing.

4.3.2.1.4.1 Examination of closure, waterproofing and banding. When shipping containers are required to comply with PPP-B-636 or PPP-P-704, examination for defects in closure, waterproofing and banding shall be in accordance with the appendix of that specification.

4.4 Test methods.

4.4.1 pH value. Make a 1.0 percent solution of the product in freshly boiled and cooled (room temperature) distilled water, as specified in 3.4, and mix. Determine the pH value of this solution to the nearest 0.1 unit at room temperature (23° ± 3°C) by means of a suitable pH meter which employs the glass electrode. No correction for sodium ion concentration is necessary.

4.4.2 Solubility in hard water. To 10 ml of the product in a 100 ml graduated cylinder, add 90 ml of 300 ppm hard water (see 4.4.2.1). Stopper and shake vigorously for 5 minutes. Allow to stand at room temperature for one hour. There shall be no separation of the emulsion.

4.4.2.1 Synthetic hard water (300 ppm hardness). To a 1 liter volumetric flask, add 0.264 g calcium chloride dihydrate ($CaCl_2 \cdot 2H_2O$) and 0.295 g magnesium sulfate heptahydrate ($MgSO_4 \cdot 7H_2O$). Add a small portion of distilled or dionized water to dissolve the salts; then dilute the solution to the mark.

4.4.3 Rinsing properties. Prepare a 2 percent solution by adding 2.0 ml of product to 98 ml of synthetic hard water (see 4.4.2.1) in a clean 250 ml Erlenmeyer flask at room temperature (23° ± 3°C). Stopper the flask and shake vigorously for 1 minute. Pour out the solution. Rinse the flask by the same procedure, using three 75 ml portions of synthetic hard water alone. Invert the flask, allow to dry, and examine for any residue not rinsed from the interior. The flask shall contain no more residue after drying than a similar flask allowed to dry after rinsing with synthetic hard water alone.

4.4.4 Cleaning efficiency. The cleaning efficiency shall be determined in accordance with Fed. Test Method Std. No. 536/6701-T with the following exception: Paragraph 1.2 under Apparatus and Material, delete "amber" and substitute "green" for the filter.

4.4.5 Composition.

4.4.5.1 Total solids. Weigh a 10 g sample to the nearest mg in a tared weighing dish. Dry the sample in an air-circulating oven at 105° ± 2°C for four hours (minimum) and to a constant weight. Reserve for 4.4.5.2.

$$\text{Total solids, percent} = \frac{\text{residue weight}}{\text{sample weight}} \times 100$$

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4.4.5.2 Synthetic organic detergent. The residue from 4.4.5.1 shall be used. The synthetic detergent content shall be determined in accordance with ASTM D 820, Sections 13 to 14, as applicable.

$$\text{Synthetic detergent, percent} = \frac{\text{weight of alcohol-soluble matter}}{\text{sample weight (see 4.4.5.1)}} \times 100$$

4.4.5.3 Water. The water content of the product shall be determined in accordance with ASTM D 95, using xylene as the solvent-carrier liquid. The sample (10-100 g, depending on the estimated water content) shall be accurately weighed in a tared 500 ml round bottom flask. Boiling chips and 250 ml of xylene shall be used.

$$\text{Water, percent} = \frac{\text{volume of water in trap}}{\text{weight of sample}} \times 100$$

4.4.5.4 Volatile solvents. The volatile solvents content shall be determined in accordance with ASTM D 322, using 10 ml of product, 250 ml of distilled or deionized water, and a 500 ml flask.

$$\text{Volatile solvents, percent} = \frac{\text{volume of solvent in trap}}{\text{volume of product used}} \times 100$$

4.4.6 Deleterious action on painted surfaces.

4.4.6.1 Measurement of specular gloss. Gloss shall be determined in accordance with ASTM D 523.

4.4.6.2 Test panels. Stainless steel (QQ-S-766) approximately 3-1/2 inches by 6 inches by 1/16 inch shall be used. The panel shall be prepared by applying (brush or spray) one coat of each of the following paints: (1) enamel undercoat, white (TT-E-543), as the first coat. (2) enamel, semi-gloss, white, (TT-E-508), with a 60° gloss of 50 to 60 as the final coat. A drying time of not less than 24 hours between coats, and a minimum of not less than 24 hours after the application of the final coat shall be allowed. When tests are to be made on panels that had been used, roughen the surface (to obtain good adhesion) before applying the final coat, (TT-E-508, 60° gloss 50-60, white). Allow the final coat to dry for 1 day minimum or 4 days maximum.

4.4.6.3 Procedure. Prepare one liter each of a solution of trisodium phosphate (Sodium phosphate, tribasic) conforming to type II of O-S-642 and a solution of the cleaning compound under test at concentrations of 2 grams per liter in distilled water. Measure the specular gloss of two panels and then immerse one panel in each of these solutions, kept at room temperature, for one hour. Remove the panels, rinse thoroughly in distilled water, and dry at room temperature for at least one hour. Again measure the specular gloss of each panel. The percentage decrease in gloss of the portion of the panel immersed, compared with the gloss of the same panel before immersion, shall be reported for the compound under test and for the trisodium phosphate.

4.4.7 Deleterious action on linoleum, rubber, and vinyl-asbestos tile flooring.

4.4.7.1 Test panels. These panels shall be as specified below:

White linoleum panels - 2 by 4 by 1/8 inches, conforming to LLL-F-1238, type I, class 2, grade A.
Black rubber tile panels - 2 by 4 1/8 inches, conforming to SS-T-312, type II.
Black vinyl-asbestos tile panels, - 2 by 4 by 1/8 inches, conforming to SS-T-312, type IV.

4.4.7.2 Procedure. Pour 150 ml of the cleaning compound solution at the concentrations specified in 3.10, into each of three 250 ml beakers. Immerse one panel of each type in the beakers containing the cleaning solution. Allow the panels to remain in the solution at room temperature (23° ± 3°C), for 16 hours. At the end of this time, remove the panels, rinse thoroughly with distilled water, and dry at room temperature for at least one hour. Examine for softening, swelling, cracking, bleaching and discoloration.

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4.4.8 Biodegradability of the synthetic detergents.

4.4.8.1 Anionic synthetic detergents. The supplier shall submit a certificate of compliance with the requirements for biodegradability. The certificate shall be accompanied by actual test date (field or laboratory), including the test procedure utilized in making the biodegradability determination. When the detergent is an alkyl benzene sulfonate or a linear alkylate sulfonate, the biodegradability shall be determined in accordance with the Test Procedure and Standards ABS and LAS Biodegradability Scientific and Technical Report No. 3 of the Soap and Detergent Association.

4.4.8.2 Nonionic and other synthetic detergents. Until an acceptable method of determining the biodegradability of a nonionic detergent is agreed upon by industry and is acceptable to the Government, the supplier shall certify that his product meets the 90 percent biodegradability requirement when tested in accordance with one of the current test methods. The certificate shall be accompanied by actual test date (field or laboratory report), indicating test method used in making the biodegradability determination.

4.4.9 Flash point. The flash point shall be determined in accordance with ASTM D 1310.

5. PREPARATION FOR DELIVERY.

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A and B.

5.1.1.1 One gallon quantity. One gallon of detergent shall be packaged in a can conforming to PPP-C-96, type V, class 4 or 5 or in a plastic container of high density polyethylene sealed with a suitable closure to prevent leakage.

5.1.1.2 Five gallon quantity.

5.1.1.2.1 Detergent shall be packaged in a pail conforming to PPP-P-704, type I or II, class 8. The pail shall be closed in accordance with the appendix to the pail specification.

5.1.1.2.2 Detergent shall be packaged in a pail conforming to PPP-P-1655, type II. The pail shall be closed to prevent leakage.

5.1.2 Level C. Detergent shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving agency.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 One gallon quantity. Six (6) one gallon containers shall be packed in a box conforming to PPP-B-636, class weather-resistant. The box shall be provided with cells, partitions, dividers or separators fabricated of fiberboard conforming to PPP-F-320, grade not less than 200. A box style designed to provide separation of containers, as an RSC with long flaps, is acceptable. The box shall be in accordance with the appendix to the box specification.

5.2.1.2 Five gallon quantity. The detergent packaged as specified in 5.1 does not require packing.

5.2.2 Level B. Detergent packaged as specified in 5.1 shall be packed as specified in 5.2.1.1 except the box shall be of fiberboard conforming to class domestic.

5.2.3 Level C. Detergent in quantities as specified (see 6.2), packaged as specified in 5.1, shall be packed in containers to assure carrier acceptance and safe arrival at destination. Containers shall comply with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special marking required in the procurement document and 3.13, marking of the interior packages and exterior shipping containers shall be in accordance with Fed. Std. No. 123 for civil agencies or MIL-STD-129 for military agencies, as applicable (see 6.2).

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6. NOTES

6.1 Intended use. The solvent-type liquid detergent covered by this specification is intended for heavy duty cleaning operations (e.g., floors, walls, concrete, general maintenance, etc.) where grease, oil, sludge, and carbonaceous deposits are encountered. The product contains organic solvents. Follow directions shown on the label. Use the product with adequate ventilation and do not breathe the vapors.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size of the container required (see 5.1).
(c) Quantity required.
(d) Levels of packaging, and packing required (see 5.1 and 5.2).
(e) Marking required (see 5.3).

MILITARY INTERESTS:

DOD has waived coordination on revisions and amendments to this Federal Specification until further notice.

User activity:

Army - GL

CIVIL AGENCY COORDINATING ACTIVITIES:

GPO
GSA-FSS
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PREPARING ACTIVITY: GSA-FSS

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